

Chapter 05 Part 3: Composition Analysis

1. A student finds the mass of a pure sample of a metal chloride hydrate. The student heats the sample several times, reaching a constant mass that contains 0.100 mole of anhydrous metal chloride. Which of the following questions about the hydrated metal chloride substance is most likely to be answered by the results of the experiment?

(A) How many moles of water are in one mole of the substance? ✓

(B) How many moles of the metal cation are in one mole of the substance?

(C) What is the identity of the metal cation in the formula of the substance?

(D) What is the melting point of the substance?

2. Two different ionic compounds each contain only copper and chlorine. Both compounds are powders, one white and one brown. An elemental analysis is performed on each powder. Which of the following questions about the compounds is most likely to be answered by the results of the analysis?

(A) What is the density of each pure compound?

(B) What is the formula unit of each compound? ✓

(C) What is the chemical reactivity of each compound?

(D) Which of the two compounds is more soluble in water?

3. To gravimetrically analyze the silver content of a piece of jewelry made from an alloy of Ag and Cu, a student dissolves a small preweighed sample in $\text{HNO}_3(\text{aq})$. $\text{Ag}^+(\text{aq})$ and $\text{Cu}^{2+}(\text{aq})$ ions form in the solution. Which of the following should be the next step in the analytical process?

(A) Centrifuging the solution to isolate the heavier ions

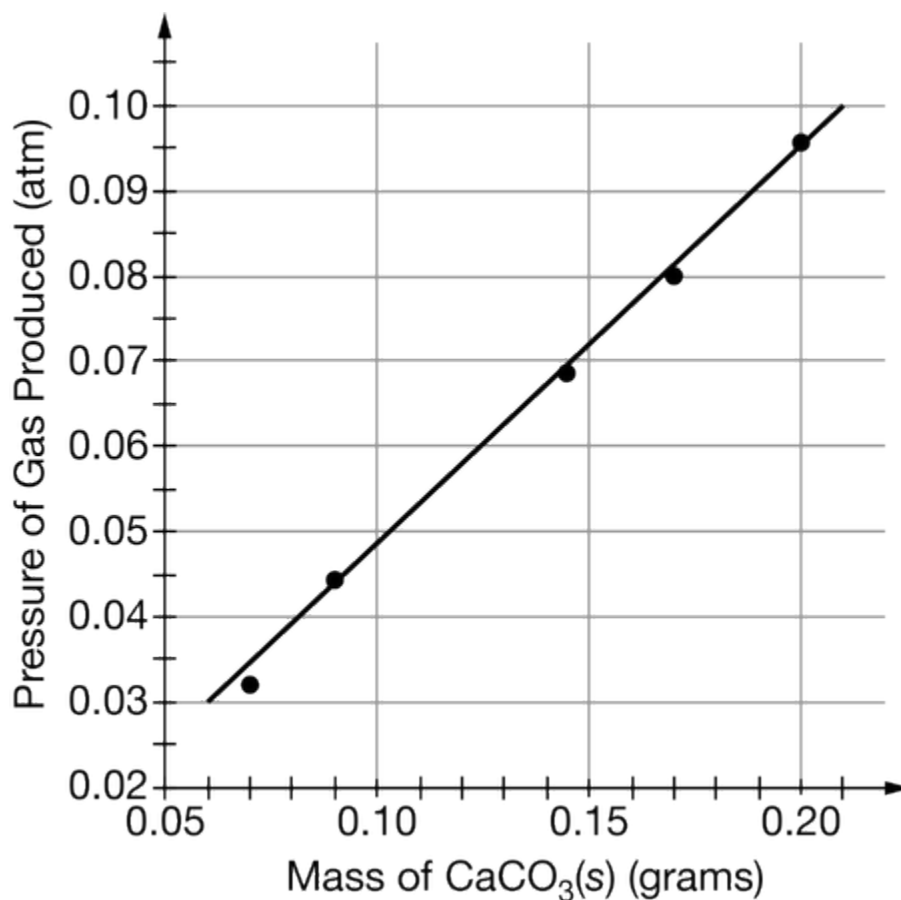
(B) Evaporating the solution to recover the dissolved nitrates

(C) Adding enough base solution to bring the pH up to 7.0

(D) Adding a solution containing an anion that forms an insoluble salt with only one of the metal ions ✓

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An experiment is performed to measure the mass percent of $\text{CaCO}_3(s)$ in eggshells. Five different samples of $\text{CaCO}_3(s)$ of known mass react with an excess of $2.0\text{ M HCl}(aq)$ in identical sealed, rigid reaction vessels. The pressure of the gas produced is measured with a pressure gauge attached to the reaction vessel. Since the reaction is exothermic, the reaction system is cooled to its original temperature before the pressure is recorded. The experimental data are used to create the calibration line below.



The experiment is repeated with an eggshell sample, and the experimental data are recorded in the table below.

Mass of eggshell sample	0.200 g
Pressure prior to reaction	0.800 atm
Pressure at completion of reaction	0.870 atm

4. The mass percent of $\text{CaCO}_3(s)$ in the eggshell sample is closest to

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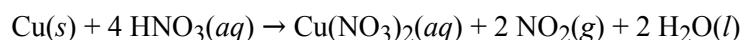
(A) 30%

(B) 45%

(C) 60%

(D) 75% 

5. In which of the following compounds is the mass ratio of chromium to oxygen closest to 1.62 to 1.00?

(A) CrO_3 (B) CrO_2 (C) CrO (D) Cr_2O (E) Cr_2O_3 

Each student in a class placed a 2.00 g sample of a mixture of Cu and Al in a beaker and placed the beaker in a fume hood. The students slowly poured 15.0 mL of 15.8 M $\text{HNO}_3(aq)$ into their beakers. The reaction between the copper in the mixture and the $\text{HNO}_3(aq)$ is represented by the equation above. The students observed that a brown gas was released from the beakers and that the solutions turned blue, indicating the formation of $\text{Cu}^{2+}(aq)$. The solutions were then diluted with distilled water to known volumes.

6. The students determined that the reaction produced 0.010 mol of $\text{Cu}(\text{NO}_3)_2$. Based on the measurement, what was the percent of Cu by mass in the original 2.00 g sample of the mixture?

(A) 16%

(B) 32% 

(C) 64%

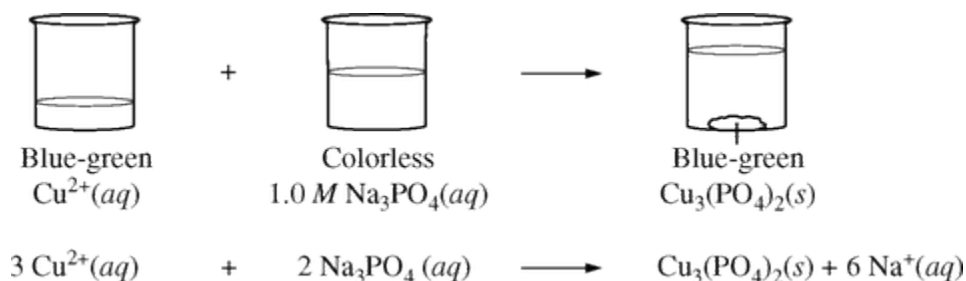
(D) 96%

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7.

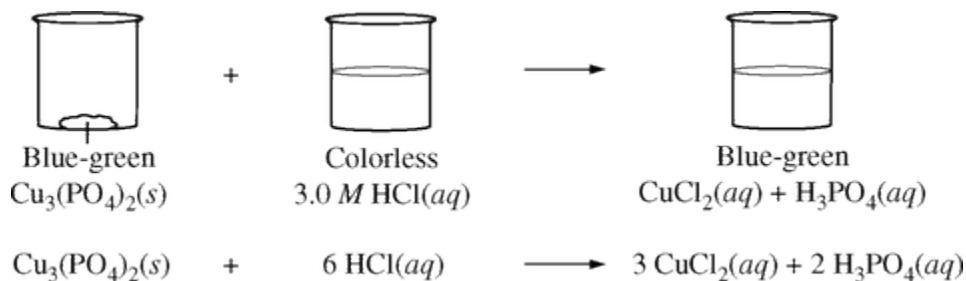
A group of students was asked to recover $\text{Cu}(s)$ from a blue-green aqueous solution containing an unknown concentration of $\text{Cu}^{2+}(aq)$. The students took a 100.0 mL sample of the solution and added an excess of 1.0 M $\text{Na}_3\text{PO}_4(aq)$, causing the $\text{Cu}^{2+}(aq)$ to precipitate as $\text{Cu}_3(\text{PO}_4)_2(s)$, as shown in step 1 below.

Step 1



The $\text{Cu}_3(\text{PO}_4)_2(s)$ was filtered, dried, and weighed. Then the $\text{Cu}_3(\text{PO}_4)_2(s)$ was dissolved in a 3.0 M $\text{HCl}(aq)$ solution, as shown in step 2 below.

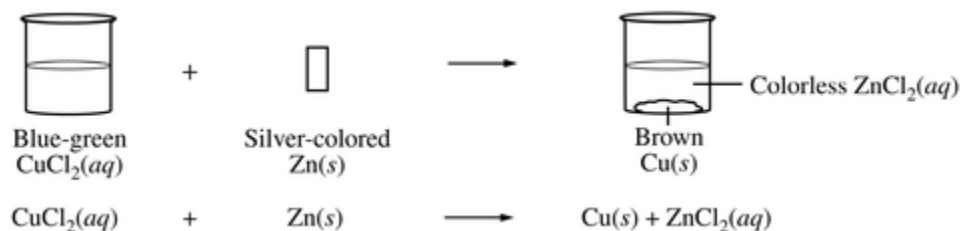
Step 2



The students added a strip of $\text{Zn}(s)$ to the solution to recover the $\text{Cu}(s)$, as shown in step 3 below.

Step 3

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Finally, the $\text{Cu}(s)$ was filtered, dried, and weighed.

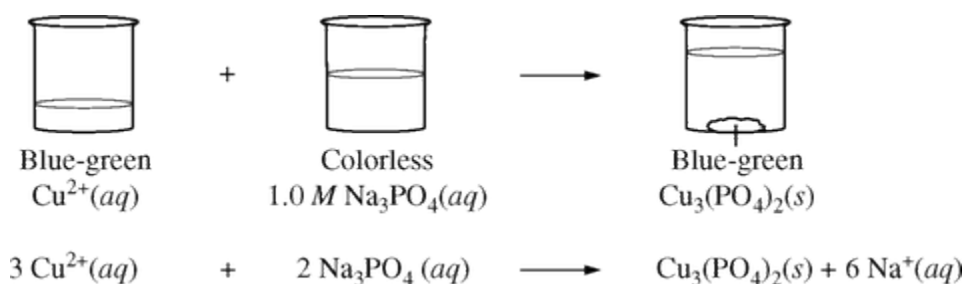
The mass of the $\text{Cu}(s)$ produced in step 3 was slightly more than the mass predicted from the 4.6 g of $\text{Cu}_3(\text{PO}_4)_2(s)$ recovered from step 1. Which of the following could account for the discrepancy in the yield of $\text{Cu}(s)$ from step 3 ?

- (A) All of the $\text{Cu}_3(\text{PO}_4)_2(s)$ dissolved in the $\text{HCl}(aq)$.
- (B) Some of the $\text{Cu}(s)$ adhered to the side of the funnel used to filter the solution.
- (C) Some unreacted $\text{Zn}(s)$ was mixed in with the $\text{Cu}(s)$. ✓
- (D) Too much $\text{Na}_3\text{PO}_4(aq)$ was added to the original $\text{Cu}^{2+}(aq)$ solution.

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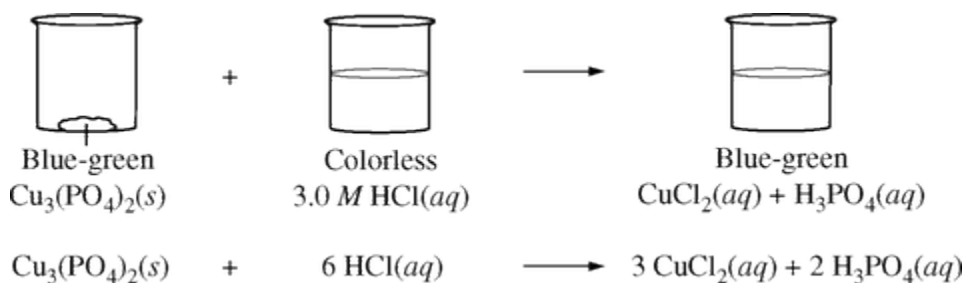
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Step 1



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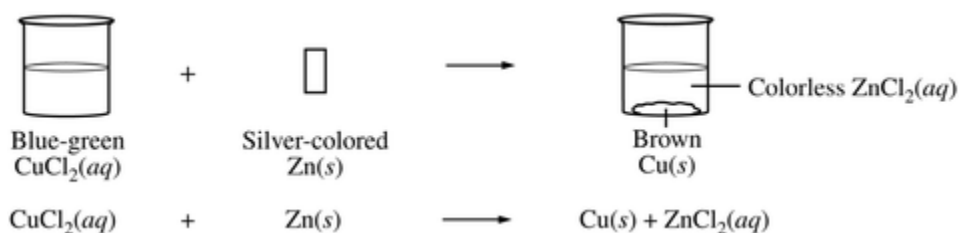
Step 2



The students added a strip of $\text{Zn}(s)$ to the solution to recover the $\text{Cu}(s)$, as shown in step 3 below.

Step 3

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Finally, the $\text{Cu}(s)$ was filtered, dried, and weighed.

8. The mass of the $\text{Cu}(s)$ produced in step 3 was slightly more than the mass predicted from the 3.8 g of $\text{Cu}_3(\text{PO}_4)_2(s)$ recovered from step 1. Which of the following could account for the discrepancy in the yield of $\text{Cu}(s)$ from step 3?
- (A) All of the $\text{Cu}_3(\text{PO}_4)_2(s)$ dissolved in the $\text{HCl}(aq)$.
- (B) Some of the $\text{Cu}(s)$ adhered to the side of the funnel used to filter the solution.
- (C) Some unreacted $\text{Zn}(s)$ was mixed in with the $\text{Cu}(s)$. ✓
- (D) Too much $\text{Na}_3\text{PO}_4(aq)$ was added to the original $\text{Cu}^{2+}(aq)$ solution.
-
9. A sample of a compound that contains only the elements C, H, and N is completely burned in O_2 to produce 44.0 g of CO_2 , 45.0 g of H_2O , and some NO_2 . A possible empirical formula of the compound is
- (A) CH_2N
- (B) CH_5N ✓
- (C) $\text{C}_2\text{H}_5\text{N}$
- (D) $\text{C}_3\text{H}_3\text{N}_2$
10. A 23.0g sample of a compound contains 12.0g of C, 3.0g of H, and 8.0g of O. Which of the following is the empirical formula of the compound?
- (A) CH_3O
- (B) $\text{C}_2\text{H}_6\text{O}$ ✓
- (C) $\text{C}_3\text{H}_9\text{O}_2$
- (D) $\text{C}_4\text{H}_{12}\text{O}_2$
11. A sample of a compound contains 3.21 g of sulfur and 11.4 g of fluorine. Which of the following represents the empirical formula of the compound?

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- (A) SF_2
- (B) SF_3
- (C) SF_4
- (D) SF_5

(E) SF_6



12. What is the empirical formula of an oxide of chromium that is 48 percent oxygen by mass?

- (A) CrO
- (B) CrO_2

(C) CrO_3



- (D) Cr_2O
- (E) Cr_2O_3

13. When hafnium metal is heated in an atmosphere of chlorine gas, the product of the reaction is found to contain 62.2 percent Hf by mass and 37.4 percent Cl by mass. What is the empirical formula for this compound?

- (A) HfCl
- (B) HfCl_2

(C) HfCl_3



- (D) HfCl_4
- (E) Hf_2Cl_3

14. A compound contains 1.10 mol of K, 0.55 mol of Te, and 1.65 mol of O. What is the simplest formula of this compound?

- (A) KTeO
- (B) KTe_2O

(C) K_2TeO_3



- (D) K_2TeO_6
- (E) K_4TeO_6

15. A compound contains 30. percent sulfur and 70. percent fluorine by mass. The empirical formula of the compound is

- (A) SF
- (B) SF_2

(C) SF_4



- (D) SF_6
- (E) S_2F

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16. If a pure sample of an oxide of sulfur contains 40. percent sulfur and 60. percent oxygen by mass, then the empirical formula of the oxide is

(A) SO_3



(B) SO_4

(C) S_2O_6

(D) S_2O_8

17. After completing an experiment to determine gravimetrically the percentage of water in a hydrate, a student reported a value of 38 percent. The correct value for the percentage of water in the hydrate is 51 percent. Which of the following is the most likely explanation for this difference?

(A) Strong initial heating caused some of the hydrate sample to spatter out of the crucible.

(B) The dehydrated sample absorbed moisture after heating.



(C) The amount of the hydrate sample used was too small.

(D) The crucible was not heated to constant mass before use.

(E) Excess heating caused the dehydrated sample to decompose.

18. The percentage of silver in a solid sample is determined gravimetrically by converting the silver to $\text{Ag}^+(aq)$ and precipitating it as silver chloride. Failure to do which of the following could cause errors in the analysis?

I. Account for the mass of the weighing paper when determining the mass of the sample

II. Measure the temperature during the precipitation reaction

III. Wash the precipitate

IV. Heat the AgCl precipitate to constant mass

(A) I only

(B) I and II

(C) I and IV

(D) II and III

(E) I, III, and IV



19. A sample of a pure compound is analyzed and found to contain approximately 30 percent N and 70 percent O by mass. The formula for the compound could be

(A) NO

(B) NO_2



(C) N_2O

(D) N_2O_2

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20. Complete combustion of a sample of a hydrocarbon in excess oxygen produces equimolar quantities of carbon dioxide and water. Which of the following could be the molecular formula of the compound?

(A) C_2H_2

(B) C_2H_6

(C) C_4H_8



(D) C_6H_6

21.

Sample	Mass of Carbon	Mass of Hydrogen
A	60. g	12 g
B	72 g	12 g
C	84 g	10. g
D	90. g	10. g

The masses of carbon and hydrogen in samples of four pure hydrocarbons are given above. The hydrocarbon in which sample has the same empirical formula as propene, C_3H_6 ?

(A) Sample A

(B) Sample B



(C) Sample C

(D) Sample D

22. A sample of a solid labeled as NaCl may be impure. A student analyzes the sample and determines that it contains 75 percent chlorine by mass. Pure $\text{NaCl}(s)$ contains 61 percent chlorine by mass. Which of the following statements is consistent with the data?

(A) The sample contains only $\text{NaCl}(s)$.

(B) The sample contains $\text{NaCl}(s)$ and $\text{NaI}(s)$.

(C) The sample contains $\text{NaCl}(s)$ and $\text{KCl}(s)$.

(D) The sample contains $\text{NaCl}(s)$ and $\text{LiCl}(s)$.



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23. The mass percent of carbon in pure glucose, $C_6H_{12}O_6$, is 40.0 percent. A chemist analyzes an impure sample of glucose and determines that the mass percent of carbon is 38.2 percent. Which of the following impurities could account for the low mass percent of carbon in the sample?
- (A) Water, H_2O ✓
- (B) Ribose, $C_5H_{10}O_5$
- (C) Fructose, $C_6H_{12}O_6$, an isomer of glucose
- (D) Sucrose, $C_{12}H_{22}O_{11}$
24. To determine the percentage of water in a hydrated salt, a student heated a 1.2346 g sample of the salt for 30 minutes; when cooled to room temperature, the sample weighed 1.1857 g. After the sample was heated for an additional 10 minutes and again cooled to room temperature, the sample weighed 1.1632 g. Which of the following should the student do next?
- (A) Use the smallest mass value to calculate the percentage of water in the hydrated salt.
- (B) Repeat the experiment with a new sample of the same mass and average the results.
- (C) Repeat the experiment with a new sample that has a different mass.
- (D) Reheat the sample until its mass is constant. ✓
- (E) Use the average of the mass values obtained after the two heatings to calculate the percentage of water in the hydrated salt.

The following questions refer to the below.

M^+ is an unknown metal cation with a +1 charge. A student dissolves the chloride of the unknown metal, MCl , in enough water to make 100.0 mL of solution. The student then mixes the solution with excess $AgNO_3$ solution, causing $AgCl$ to precipitate. The student collects the precipitate by filtration, dries it, and records the data shown below. (The molar mass of $AgCl$ is 143 g/mol.)

Mass of unknown chloride, MCl	0.74 g
Mass of filter paper	0.80 g
Mass of filter paper plus $AgCl$ precipitate	2.23 g

25. What is the identity of the metal chloride?
- (A) $NaCl$
- (B) KCl ✓
- (C) $CuCl$
- (D) $LiCl$

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26. A student is given two 10g samples, each a mixture of only $\text{NaCl}(s)$ and $\text{KCl}(s)$ but in different proportions. Which of the following pieces of information could be used to determine which mixture has the higher proportion of $\text{KCl}(s)$?

(A) The volume of each mixture

(B) The mass of Cl in each mixture



(C) The number of isotopes of Na and K

(D) The reaction of each mixture with water

27.

1	2	3	4	5	6	7
Total mass of sample (g)	Mass of C in sample (g)	Mass of Mg in sample (g)	Mass of Ca in sample (g)	Molar mass of C (g/mol)	Molar mass of Mg (g/mol)	Molar mass of Ca (g/mol)
98.5	12.0	2.4	36.1	12.0	24.3	40.1

A sample of carbonate rock is a mixture of CaCO_3 and MgCO_3 . The rock is analyzed in a laboratory, and the results are recorded in the table above. Which columns in the table provide all the information necessary to determine the mole ratio of Ca to Mg in the rock?

(A) 1, 2, 5

(B) 2, 5, 6

(C) 3, 4, 6, 7



(D) 2, 3, 4, 5

28. A student has two samples of NaCl , each one from a different source. Assume that the only potential contaminant in each sample is KCl . The student runs an experiment to determine the percent by mass of chlorine in each sample. From the results of this experiment alone, which of the following questions is most likely to be answered?

(A) Which sample has the higher purity?



(B) Which sample has the higher density?

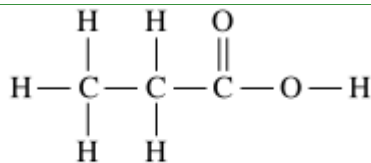
(C) What is the source of the contaminants present in each of the samples?

(D) Which sample came from a salt mine, and which sample came from the ocean?

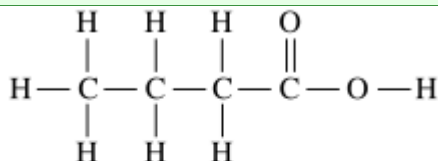
29. A student determines that an unknown compound contains 49 percent C by mass. Which of the following could be the compound?

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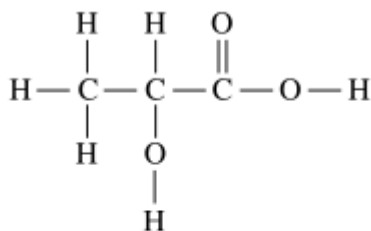
(A)

Propanoic acid
(74 g/mol)

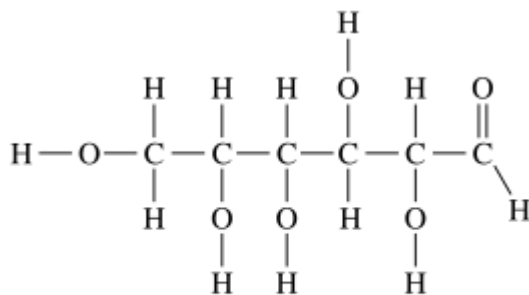
(B)

Butanoic acid
(88 g/mol)

(C)

Lactic acid
(90 g/mol)

(D)

Glucose
(180 g/mol)

30. A sample of CaCO_3 (molar mass 100. g) was reported as being 30. percent Ca. Assuming no calcium was present in any impurities, the percent of CaCO_3 in the sample is

(A) 30%

(B) 40%

(C) 70%

(D) 75%

(E) 100%



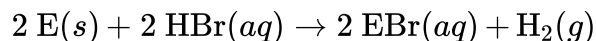
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31. A 5.0 g sample of MgCl_2 may contain measurable amounts of other compounds as impurities. Which of the following quantities is (are) needed to determine that the sample is pure MgCl_2 ?
- (A) The color and density of the sample
 - (B) The mass of Mg in the sample only
 - (C) The number of moles of Cl in the sample only
 - (D) The mass of Mg and the mass of Cl in the sample 
32. A student has samples of two pure compounds, XClO_3 and ZClO_3 , which contain unknown alkali metals X and Z. The student measures the mass of each sample and then strongly heats the samples to drive off all the oxygen, leaving solid residues of XCl and ZCl . The student measures the mass of the solid residue from each sample. Which of the following questions can be answered from the results of the experiment?
- (A) Which has the greater molar mass, X or Z? 
 - (B) Which has the higher boiling point, X or Z?
 - (C) Which has the higher melting point, XCl or ZCl ?
 - (D) Which has the greater density, XCl or ZCl ?
33. When a 3.22 g sample of an unknown hydrate of sodium sulfate, $\text{Na}_2\text{SO}_4 \cdot \text{H}_2\text{O}(s)$, is heated, H_2O (molar mass 18 g) is driven off. The mass of the anhydrous $\text{Na}_2\text{SO}_4(s)$ (molar mass 142 g) that remains is 1.42 g. The value of x in the hydrate is
- (A) 0.013
 - (B) 1.8
 - (C) 6.0
 - (D) 10. 
 - (E) 20.

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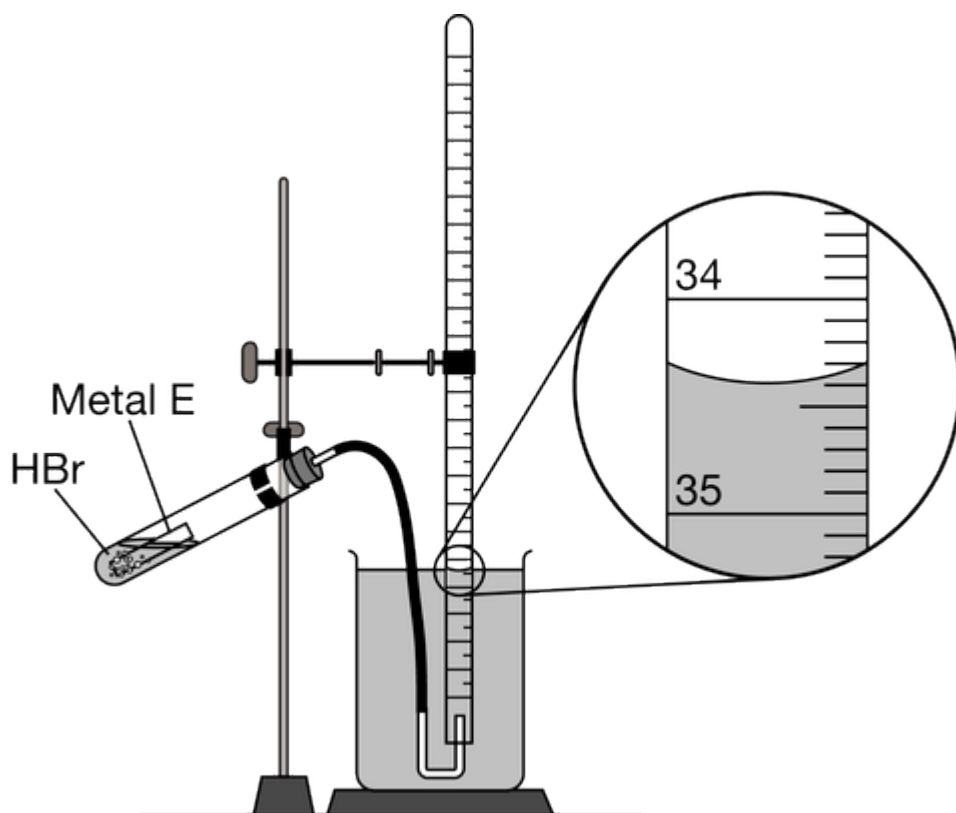
34. Directions: For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

While investigating the chemical and physical properties of metallic element E, a student conducts an experiment by combining E(s) and HBr(aq), which react according to the following equation.



- (a) Write the net-ionic equation for the reaction.
- (b) In the reaction, is E(s) being oxidized or reduced? Justify your answer in terms of oxidation numbers.

As shown in the following diagram, 10.00 mL samples of HBr(aq) are added to different quantities of E(s) in a closed test tube. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of E(s) react with HBr(aq), and the H₂(g) produced is collected in the gas-collecting tube.



- (c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in mL, of H₂(g) collected in trial 3?

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(d) In the laboratory, the temperature is 295 K and the total pressure in the gas-collecting tube is 748.6 mmHg. If the vapor pressure of water at 295 K is 19.8 mmHg, determine the pressure of the $\text{H}_2(g)$ in the gas-collecting tube.

Trial	Initial Moles of $\text{E}(s)$	Initial $[\text{HBr}(aq)]$ (M)	Volume of $\text{H}_2(g)$ Produced (mL)
1	0.00220	0.500	27.02
2	0.00240	0.500	29.40
3	0.00280	0.500	?

(e) The data from all three trials are given above. A student claims that $\text{HBr}(aq)$ is the limiting reactant in trial 1. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element E is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

Element	Moles Present in the Compound
E	0.0676
S	0.0338
O	0.102

(g) A different element, L , is located in the same row of the periodic table as element E , but L has roughly half the atomic mass of E . Is the atomic radius of L less than, equal to, or greater than the atomic radius of E ? Justify your answer using principles of atomic structure.

(h) The ionic radii of elements E and a different metallic element, M , are shown in the following table. Both elements form oxides, E_2O and MO . If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of MO be less than, equal to, or greater than the lattice energy of the oxide E_2O ? Justify your answer in terms of Coulomb's law.

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Element	Ionic radius (pm)
E	170
M	110

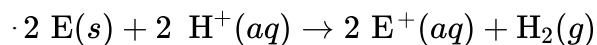
Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct net ionic equation (state symbols not required).



Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.



Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows the correct volume.

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· 34.39 mL

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct pressure of $\text{H}_2(g)$.

· $748.6 \text{ mmHg} - 19.8 \text{ mmHg} = 728.8 \text{ mmHg}$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to one of the following.

· Disagree. Given the 1 : 1 reaction stoichiometry between E and HBr , the 0.00500 mol of HBr exceeds the 0.00220 mol of E that is present, so HBr is in excess.

OR

· Disagree. More H_2 is produced when the initial amount of E is increased and the amount of HBr is held constant. This result is consistent with E being the limiting reactant.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct empirical formula and a mathematical or written justification equivalent to the following.

· E_2SO_3

$$\text{E} = 0.0676 \text{ mol} / 0.0338 \text{ mol} = 2$$

$$\text{S} = 0.0338 \text{ mol} / 0.0338 \text{ mol} = 1$$

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$$\text{O} = 0.102 \text{ mol} / 0.0338 \text{ mol} = 3$$

OR

· E_2SO_3 . There are twice as many moles of E as there are of S, and three times as many moles of O as there are of S.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

· Greater than. Since both elements are in the same row of the periodic table, their valence electrons occupy the same shell. However, since L has a substantially lower atomic mass than E, then L must also have a lower atomic number and therefore fewer protons in its nucleus. The valence electrons in L thus experience a weaker effective nuclear charge compared to E and are, on average, further from the nucleus because they are less strongly attracted.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to one of the following.

· Greater than. According to Coulomb's law, the strength of interaction between two ions is directly proportional to the magnitude of their ionic charges. Thus, the attraction between the metal cation and the O^{2-} charge anion is stronger in MO (M^{2+}) than in E_2O (E^+), resulting in a greater lattice energy for MO.

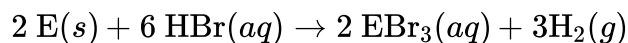
OR

· Greater than. According to Coulomb's law, the strength of interaction between two ions is inversely proportional to the square of the distance between their centers. Because the distance between the metal cation and the O^{2-} anion is shorter in MO than in E_2O , MO has the greater lattice energy.

Chapter 05 Part 3: Composition Analysis

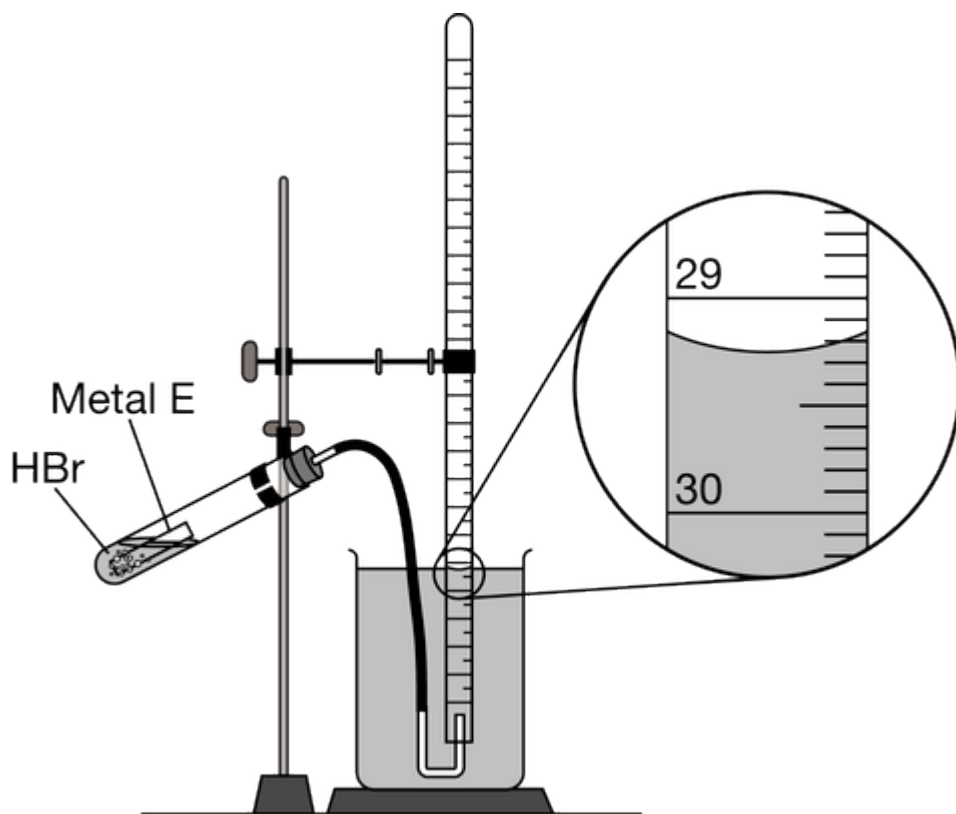
35. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

While investigating the chemical and physical properties of metallic element E, a student conducts an experiment by combining E(s) and HBr(aq), which react according to the following equation.



- (a) Write the net-ionic equation for the reaction.
- (b) In the reaction, is E(s) being oxidized or reduced? Justify your answer in terms of oxidation numbers.

As shown in the following diagram, 10.00 mL samples of HBr(aq) are added to different quantities of E(s) in a closed test tube. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of E(s) react with HBr(aq), and the H₂(g) produced is collected in the gas-collecting tube.



- (c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in mL, of H₂(g) collected in trial 3?
- (d) In the laboratory, the temperature is 295 K and the total pressure in the gas-collecting tube is 744.7 mmHg. If the vapor pressure of water at 295 K is 19.8 mmHg, determine the pressure of the H₂(g) in the gas-collecting

Chapter 05 Part 3: Composition Analysis

tube.

Trial	Initial Moles of E(s)	Initial [HBr(aq)] (M)	Volume of H ₂ (g) Produced (mL)
1	5.40×10^{-4}	0.400	19.95
2	6.90×10^{-4}	0.400	25.52
3	8.20×10^{-4}	0.400	?

(e) The data from all three trials are given above. A student claims that E(s) is the limiting reactant in trial 2. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element E is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

Element	Moles Present in the Compound
E	0.0167
I	0.050
O	0.200

(g) A different element, L, is located in the same column of the periodic table as element E, but L has roughly double the atomic mass of E. Is the atomic radius of L less than, equal to, or greater than the atomic radius of E? Justify your answer using principles of atomic structure.

(h) The ionic radii of ions X^{2−} and Z^{2−} are shown in the following table. Element E can bond with X or Z to form E₂X₃ or E₂Z₃, respectively. If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of E₂X₃ be less than, equal to, or greater than the lattice energy of E₂Z₃? Justify your answer in terms of Coulomb's law.

Element	Ionic radius (pm)
X ^{2−}	120
Z ^{2−}	180

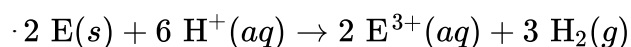
Chapter 05 Part 3: Composition Analysis**Part A**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct net ionic equation (state symbols not required).

**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

· $\text{E}(s)$ is oxidized because its oxidation number increases from 0 to +3.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response shows the correct volume.

· 29.25 mL

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct pressure of $\text{H}_2(g)$.

Chapter 05 Part 3: Composition Analysis

$$744.7 \text{ mmHg} - 19.8 \text{ mmHg} = 724.9 \text{ mmHg}$$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to one of the following.

· Agree. Given the 1 : 3 reaction stoichiometry between E and HBr, the 0.000690 mol of E is insufficient to react with the full 0.00400 mol of HBr that is present.

OR

· Agree. Relative to trial 2, in trial 1 less H₂ is produced when the initial amount of E is decreased and the amount of HBr is held constant. This result is consistent with E being the limiting reactant.

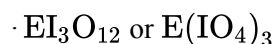
Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct empirical formula and a mathematical or written justification equivalent to one of the following.



$$\text{E} = 0.0167 \text{ mol} / 0.0167 \text{ mol} = 1$$

$$\text{I} = 0.0500 \text{ mol} / 0.0167 \text{ mol} = 3$$

$$\text{O} = 0.200 \text{ mol} / 0.0167 \text{ mol} = 12$$

OR

· EI₃O₁₂ or E(IO₄)₃. There are three times as many moles of I as there are of E, and 12 times as many moles of O as there are of E.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

· Greater than. Since L has a greater atomic mass than E, then L must also have a higher atomic number with its valence electrons occupying a higher-level shell. Electrons occupying a higher shell are, on average, further from the nucleus and thus result in a greater atomic radius.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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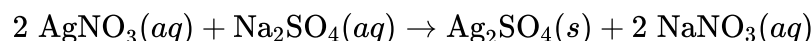
The response provides both the correct choice and a valid justification equivalent to the following.

· Greater than. According to Coulomb's law, the strength of interaction between two ions is inversely proportional to the distance between their centers. If the distance between the E^{3+} cation and the corresponding anion is shorter in E_2X_3 than in E_2Z_3 , then E_2X_3 has the greater lattice energy.

Chapter 05 Part 3: Composition Analysis

36. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

A student wishes to determine the concentration of $\text{Ag}^+(aq)$ in a solution of $\text{AgNO}_3(aq)$. The student combines 10.00 mL of $\text{AgNO}_3(aq)$ with excess $\text{Na}_2\text{SO}_4(aq)$ and observes the formation of a white precipitate. The formation of the precipitate is represented by the following equation.



- (a) Write the balanced net ionic equation for the precipitation reaction.

The student collects the precipitate by filtration and measures the mass of the filter paper and precipitate every 10 minutes as it dries. The student records the data in the following table.

Mass of dry filter paper	0.88 g
Mass of filter paper and precipitate immediately after filtration	4.82 g
Mass of filter paper and precipitate after 10 minutes	4.37 g
Mass of filter paper and precipitate after 20 minutes	4.01 g
Mass of filter paper and precipitate after 30 minutes	3.79 g

- (b) Use the data above to calculate the number of moles of $\text{Ag}_2\text{SO}_4(s)$ (molar mass 311.8 g/mol) that precipitated.
- (c) Calculate the concentration of $\text{Ag}^+(aq)$ in the original 10.00 mL solution of $\text{AgNO}_3(aq)$.
- (d) The concentration of $\text{Ag}^+(aq)$ determined by the student is significantly higher than the actual concentration of $\text{Ag}^+(aq)$. Based on the student's data table, identify an error in the experimental procedure that led to this result.

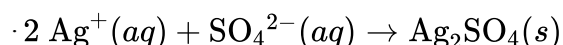
Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides a correct balanced equation (state symbols not required).



Chapter 05 Part 3: Composition Analysis**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct calculated value.

$$3.79 \text{ g} - 0.88 \text{ g} = 2.91 \text{ g}$$

$$2.91 \text{ g} \times \frac{1 \text{ mol Ag}_2\text{SO}_4}{311.8 \text{ g}} = 0.00933 \text{ mol Ag}_2\text{SO}_4$$

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct calculated value.

$$0.00933 \text{ mol Ag}_2\text{SO}_4 \times \frac{2 \text{ mol Ag}^+}{1 \text{ mol Ag}_2\text{SO}_4} = 0.0187 \text{ mol Ag}^+$$

$$\frac{0.0187 \text{ mol Ag}^+}{0.0100 \text{ L}} = 1.87M$$

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides a correct source of error.

- The student did not wait for the precipitate to dry completely (reach a constant mass)

Chapter 05 Part 3: Composition Analysis

37. For parts of the free response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

Answer the following questions related to the analysis of CaBr_2 .

- (a) A student has a 10.0 g sample of CaBr_2 . Show the setup of the calculation to determine the number of moles of CaBr_2 in the sample. Include units in the setup. (You do not need to do any calculations.)
- (b) What number, in addition to the answer to part (a), is needed to determine the number of atoms of Ca in the sample?
- (c) A different student is given a 10.0 g sample labeled CaBr_2 that may contain an inert (nonreacting) impurity. Identify a quantity from the results of laboratory analysis that the student could use to determine whether the sample was pure.
- (d) Explain why CaCl_2 is likely to have properties similar to those of CaBr_2 .

Part (a)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives one of the following setups (or an equivalent variant):

- ☐ $10.0 \text{ g CaBr}_2 \times \frac{1 \text{ mol CaBr}_2}{199.88 \text{ g CaBr}_2} = \text{number of moles of CaBr}_2$ (right side of equation not necessary)
- ☐ $\frac{1 \text{ mol CaBr}_2}{199.88 \text{ g}} = \frac{x \text{ mol CaBr}_2}{10.0 \text{ g}}$
- ☐ $\frac{10.0 \text{ g CaBr}_2}{199.88 \text{ g CaBr}_2/\text{mol}}$

Note: Molar mass can be written with any number of significant figures.

Part (b)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response gives the term “Avogadro’s number” or the value of Avogadro’s number, 6.022×10^{23} (with any number

Chapter 05 Part 3: Composition Analysis

of significant figures, units of mol^{-1} or per mole are not required).

Part (c)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response does give an appropriate result (such as one of the following):

- ☐ Mass of Ca in sample
- ☐ Number of moles of Ca in sample
- ☐ Mass of Br in sample
- ☐ Number of moles of Br in sample
- ☐ Mass or number of moles of element other than Ca or Br in sample

Part (d)

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



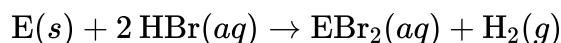
0	1
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The response indicates that Cl is in the same group/family/column of the periodic table as Br OR that a Cl atom has the same number of valence electrons as a Br atom.

Chapter 05 Part 3: Composition Analysis

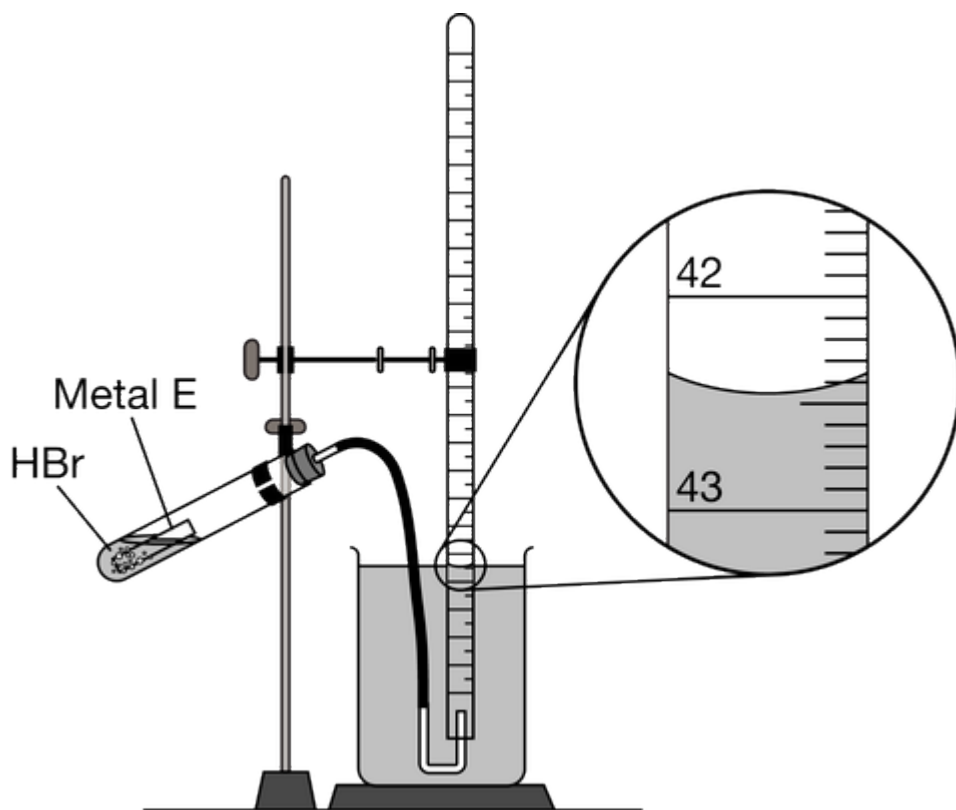
38. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

While investigating the chemical and physical properties of metallic element E, a student conducts an experiment by combining E(s) and HBr(aq), which react according to the following equation.



- (a) Write the net-ionic equation for the reaction.
- (b) In the reaction, is E(s) oxidized or reduced? Justify your answer in terms of oxidation numbers.

As shown in the following diagram, 10.00 mL samples of HBr(aq) of various concentrations are added to samples of E(s) in a closed test tube. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of E(s) react with HBr(aq), and the H₂(g) produced is collected in the gas-collecting tube.



- (c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in mL, of H₂(g) collected in trial 3?

Chapter 05 Part 3: Composition Analysis

(d) In the laboratory, the temperature is 294 K and the total pressure in the gas-collecting tube is 754.7 mmHg. If the vapor pressure of water at 294 K is 18.6 mmHg, determine the pressure of the $\text{H}_2(g)$ in the gas-collecting tube.

Trial	Initial Moles of $\text{E}(s)$	Initial $[\text{HBr}(aq)] (M)$	Volume of $\text{H}_2(g)$ Produced (mL)
1	0.00150	0.200	24.52
2	0.00150	0.250	30.95
3	0.00150	0.350	?

(e) The data from all three trials are given above. A student claims that $\text{E}(s)$ is the limiting reactant in trial 1. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element E is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

Element	Moles Present in the Compound
E	0.0652
C	0.130
O	0.260

(g) A different element, L, is located in the same row of the periodic table as element E, but L has roughly double the atomic mass of E. Is the atomic radius of L less than, equal to, or greater than the atomic radius of E? Justify your answer using principles of atomic structure.

(h) The ionic radii of elements E and a different metallic element, M, are shown in the following table. Both elements form oxides, EO and M_2O . If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of M_2O be less than, equal to, or greater than the lattice energy of the oxide EO ? Justify your answer in terms of Coulomb's law.

Element	Ionic radius (pm)
E	110
M	150

Part A

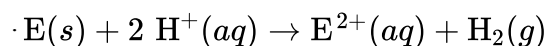
Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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Chapter 05 Part 3: Composition Analysis

The response provides the correct net ionic equation (state symbols not required).

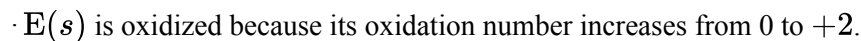
**Part B**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct answer and a valid justification equivalent to the following.

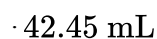
**Part C**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows the correct volume.

**Part D**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct pressure of $\text{H}_2(g)$.

**Part E**

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.

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0	1
---	---

The response provides both the correct answer and a valid justification equivalent to one of the following.

· Disagree. Given the 1 : 2 reaction stoichiometry between E and HBr, the 0.0015 mol of E will be in excess when treated with only 0.00200 mol of HBr.

OR

· Disagree. More H₂ is produced when the initial amount of HBr is increased and the amount of E is held constant. This result is consistent with HBr being the limiting reactant.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct empirical formula and a mathematical or written justification equivalent to one of the following.

· E C₂O₄

$$E = 0.0652 \text{ mol} / 0.0652 \text{ mol} = 1$$

$$C = 0.130 \text{ mol} / 0.0652 \text{ mol} = 2$$

$$O = 0.260 \text{ mol} / 0.0652 \text{ mol} = 4$$

OR

· E C₂O₄. There are twice as many moles of C as there are of E, and 4 times as many moles of O as there are of E.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

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· Less than. Since both elements are in the same row of the periodic table, their valence electrons occupy the same shell. However, since L has a substantially greater atomic mass than E, then L must also have a higher atomic number and therefore a greater number of protons in its nucleus. The valence electrons in L thus experience a greater effective nuclear charge compared to E and are, on average, closer to the nucleus because they are more strongly attracted.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to one of the following.

· Less than. According to Coulomb's law, the strength of interaction between two ions is directly proportional to the magnitude of their ionic charges. Thus, the attraction between the metal cation and the O^{2-} anion is weaker in M_2O (M^+) than in EO (E^{2+}), resulting in a lesser lattice energy for M_2O .

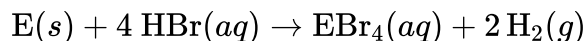
OR

· Less than. According to Coulomb's law, the strength of interaction between two ions is inversely proportional to the distance between their centers. Because the distance between the metal cation and the O^{2-} anion is greater in M_2O than in EO , M_2O has the lesser lattice energy.

Chapter 05 Part 3: Composition Analysis

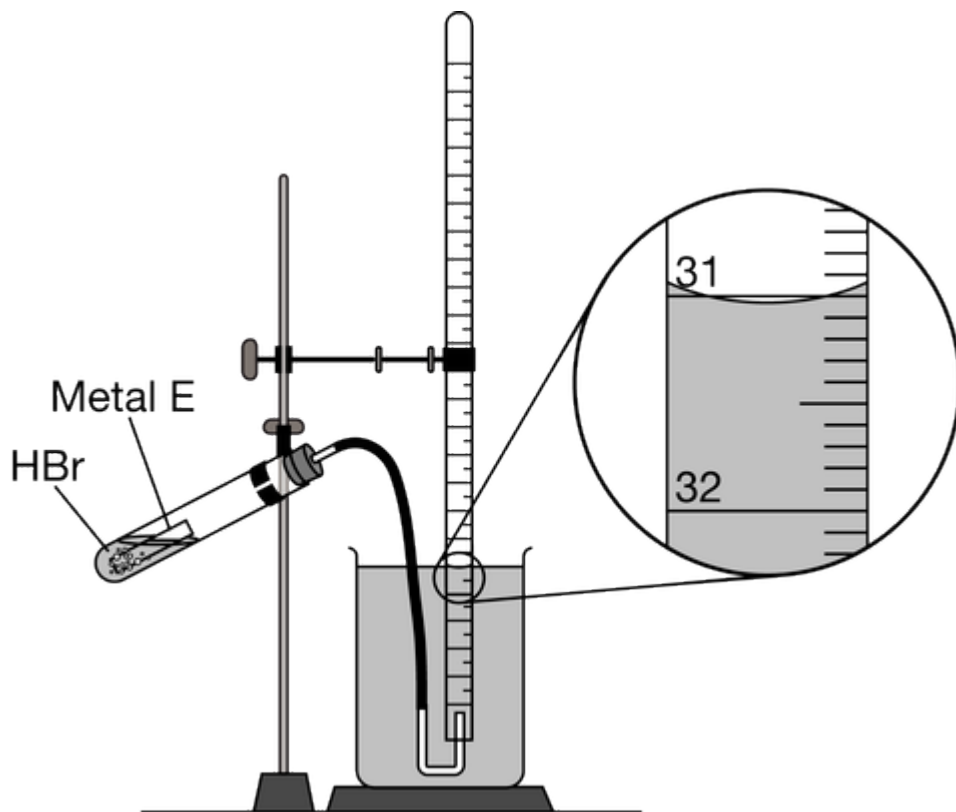
39. **Directions:** For parts of the free-response question that require calculations, clearly show the method used and the steps involved in arriving at your answers. You must show your work to receive credit for your answer. Examples and equations may be included in your answers where appropriate.

While investigating the chemical and physical properties of metallic element E, a student conducts an experiment by combining $E(s)$ and $HBr(aq)$, which react according to the following equation.



- (a) Write the net-ionic equation for the reaction.
- (b) In the reaction, is H being oxidized or reduced? Justify your answer using oxidation numbers.

As shown in the following diagram, in a closed test tube, 10.00 mL samples of $HBr(aq)$ of various concentrations are added to samples of $E(s)$. At the start of each trial, the gas-collecting tube is filled completely with distilled water. The samples of $E(s)$ react with $HBr(aq)$, and the $H_2(g)$ produced is collected in the gas-collecting tube.



- (c) The enlarged view of the gas-collecting tube at the end of trial 3 is shown in the diagram above. What should the student record as the volume, in mL, of $H_2(g)$ collected in trial 3?
- (d) In the laboratory, the temperature is 295 K and the total pressure in the gas-collecting tube is 741.2 mmHg.

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If the vapor pressure of water at 295 K is 19.8 mmHg, determine the pressure of the $\text{H}_2(g)$ in the gas-collecting tube.

Trial	Initial Moles of $\text{E}(s)$	Initial $[\text{HBr}(aq)]$ (M)	Volume of $\text{H}_2(g)$ Produced (mL)
1	7.50×10^{-4}	0.150	18.60
2	7.50×10^{-4}	0.200	24.81
3	7.50×10^{-4}	0.250	?

(e) The data from all three trials are given above. A student claims that $\text{E}(s)$ is the limiting reactant in trial 2. Do you agree or disagree? Justify your answer with information in the data table.

(f) A certain pure compound containing element E is analyzed. The elemental composition of the compound is shown in the following table. Determine the empirical formula of the compound using the information in the table.

Element	Moles Present in the Compound
E	0.0339
C	0.136
O	0.272

(g) A different element, L , is located in the same column of the periodic table as element E , but L has roughly half the atomic mass of E . Is the atomic radius of L less than, equal to, or greater than the atomic radius of E ? Justify your answer using principles of atomic structure.

(h) The ionic radii of ions X^- and Z^- are shown in the following table. Element E can bond with X or Z to form EX_4 or EZ_4 , respectively. If lattice energy is defined as the energy required to separate an ionic solid into individual separate gaseous ions, would the lattice energy of EX_4 be less than, equal to, or greater than the lattice energy of EZ_4 ? Justify your answer in terms of Coulomb's law.

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Element	Ionic radius (pm)
X^-	170
Z^-	120

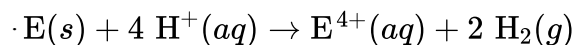
Part A

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides the correct net ionic equation (state symbols not required).



Part B

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to the following.

Hydrogen is reduced because its oxidation number decreases from +1 to 0.

Part C

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response shows the correct volume.

Chapter 05 Part 3: Composition Analysis

· 31.03 mL

Part D

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides the correct pressure of $\text{H}_2(g)$.

· $741.2 \text{ mmHg} - 19.8 \text{ mmHg} = 721.4 \text{ mmHg}$

Part E

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct answer and a valid justification equivalent to one of the following.

· Disagree. Given the 1 : 4 reaction stoichiometry between E and HBr , the 0.000750 mol of E will be in excess when reacted with only 0.00200 mol of HBr .

OR

· Disagree. Less H_2 is produced when the initial amount of HBr is decreased and the amount of E is held constant. This result is consistent with HBr , not E, being the limiting reactant.

Part F

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct empirical formula and a mathematical or written justification equivalent to one of the following.

· EC_4O_8 or $\text{E}(\text{C}_2\text{O}_4)_2$

$$\text{E} = 0.0339 \text{ mol} / 0.0339 \text{ mol} = 1$$

$$\text{C} = 0.136 \text{ mol} / 0.0339 \text{ mol} = 4$$

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$$\text{O} = 0.272 \text{ mol} / 0.0339 \text{ mol} = 8$$

OR

· EC_4O_8 or $\text{E}(\text{C}_2\text{O}_4)_2$. There are 4 times as many moles of C as there are of E, and 8 times as many moles of O as there are of E.

Part G

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
---	---

The response provides both the correct choice and a valid justification equivalent to the following.

· Less than. Since L has a lower atomic mass than E, then L must also have a lower atomic number with its valence electrons occupying a lower-level shell. Electrons occupying a lower shell are, on average, closer to the nucleus and thus result in a smaller atomic radius.

Part H

Select a point value to view scoring criteria, solutions, and/or examples and to score the response.



0	1
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The response provides both the correct choice and a valid justification equivalent to the following.

· Less than. According to Coulomb's law, the strength of interaction between two charged ions is inversely proportional to the square of the distance between their centers. If the distance between the E^{4+} cation and the corresponding anion is longer in EX_4 than in EZ_4 , then EX_4 has the lower lattice energy.

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40. Answer the following questions that relate to the analysis of chemical compounds.

- a. A compound containing the elements C, H, N, and O is analyzed. When a 1.2359 g sample is burned in excess oxygen, 2.241 g of $\text{CO}_2(\text{g})$ is formed. The combustion analysis also showed that the sample contained 0.0648 g of H.
 - i. Determine the mass, in grams, of C in the 1.2359 g sample of the compound.
 - ii. When the compound is analyzed for N content only, the mass percent of N is found to be 28.84 percent. Determine the mass, in grams, of N in the original 1.2359 g sample of the compound.
 - iii. Determine the mass, in grams, of O in the original 1.2359 g sample of the compound.
 - iv. Determine the empirical formula of the compound.
- b. A different compound, which has the empirical formula CH_2Br , has a vapor density of 6.00 g L^{-1} at 375 K and 0.983 atm. Using these data, determine the following.
 - i. The molar mass of the compound
 - ii. The molecular formula of the compound

Part A

- 1 point is earned for the correct answer.

$$2.241 \text{ g CO}_2(\text{g}) \times \frac{1 \text{ mol CO}_2}{44.01 \text{ g CO}_2} \times \frac{1 \text{ mol C}}{1 \text{ mol CO}_2} \times \frac{12.011 \text{ g C}}{1 \text{ mol C}} \\ = 0.6116 \text{ g C}$$

- 1 point is earned for the correct answer.

$$1.2359 \text{ g sample} \times 0.2884 = 0.3564 \text{ g N}$$

- 1 point is earned for the answer consistent with the answers in parts (a)(i) and (a)(ii).

Because the compound contains only C, H, N, and O, mass of O = g sample – (g H + g C + g N) = $1.2359 - (0.0648 + 0.6116 + 0.3564) = 0.2031 \text{ g}$

- 1 point is earned for all masses converted to moles.

Note: Moles of C may be shown in part (a)(i).

$$0.6116 \text{ g C} \times \frac{1 \text{ mol C}}{12.011 \text{ g C}} = 0.05092 \text{ mol C}$$

$$0.0648 \text{ g H} \times \frac{1 \text{ mol H}}{1.0079 \text{ g H}} = 0.06429 \text{ mol H}$$

$$0.3564 \text{ g N} \times \frac{1 \text{ mol N}}{14.007 \text{ g N}} = 0.02544 \text{ mol N}$$

$$0.2031 \text{ g O} \times \frac{1 \text{ mol O}}{16.00 \text{ g O}} = 0.01269 \text{ mol O}$$

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1 point is earned for dividing by the smallest number of moles.

1 point is earned for the empirical formula consistent with the ratio of moles calculated.

Divide all mole quantities by the smallest number of moles:

$$0.05092 \text{ mol} \div 0.01269 \text{ mol} = 4.013$$

$$0.06429 \text{ mol} \div 0.01269 \text{ mol} = 5.066$$

$$0.02544 \text{ mol} \div 0.01269 \text{ mol} = 2.005$$

$$0.01269 \text{ mol} \div 0.01269 \text{ mol} = 1.000$$

⇒ Empirical formula is C₄H₅N₂O



0	1	2	3	4	5	6
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The student response earns six of the following points:

- 1 point is earned for the correct answer.

$$2.241 \text{ g CO}_2(g) \times \frac{1 \text{ mol CO}_2}{44.01 \text{ g CO}_2} \times \frac{1 \text{ mol C}}{1 \text{ mol CO}_2} \times \frac{12.011 \text{ g C}}{1 \text{ mol C}}$$

$$= 0.6116 \text{ g C}$$

- 1 point is earned for the correct answer.

$$1.2359 \text{ g sample} \times 0.2884 = 0.3564 \text{ g N}$$

- 1 point is earned for the answer consistent with the answers in parts (a)(i) and (a)(ii).

Because the compound contains only C, H, N, and O, mass of O = g sample – (g H + g C + g N) = 1.2359 – (0.0648 + 0.6116 + 0.3564) = 0.2031 g

- 1 point is earned for all masses converted to moles.

Note: Moles of C may be shown in part (a)(i).

$$0.6116 \text{ g C} \times \frac{1 \text{ mol C}}{12.011 \text{ g C}} = 0.05092 \text{ mol C}$$

$$0.0648 \text{ g H} \times \frac{1 \text{ mol H}}{1.0079 \text{ g H}} = 0.06429 \text{ mol H}$$

$$0.3564 \text{ g N} \times \frac{1 \text{ mol N}}{14.007 \text{ g N}} = 0.02544 \text{ mol N}$$

$$0.2031 \text{ g O} \times \frac{1 \text{ mol O}}{16.00 \text{ g O}} = 0.01269 \text{ mol O}$$

1 point is earned for dividing by the smallest number of moles.

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1 point is earned for the empirical formula consistent with the ratio of moles calculated.

Divide all mole quantities by the smallest number of moles:

$$0.05092 \text{ mol} \div 0.01269 \text{ mol} = 4.013$$

$$0.06429 \text{ mol} \div 0.01269 \text{ mol} = 5.066$$

$$0.02544 \text{ mol} \div 0.01269 \text{ mol} = 2.005$$

$$0.01269 \text{ mol} \div 0.01269 \text{ mol} = 1.000$$

⇒ Empirical formula is C₄H₅N₂O

Part B

- 1 point is earned for applying the gas law to calculate n .

$$PV = nRT \Rightarrow \frac{PV}{RT} = n$$

$$\frac{(0.983 \text{ atm})(1.00 \text{ L})}{(0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1})(375 \text{ K})} = 0.0319$$

1 point is earned for calculating the molar mass.

$$\text{molar mass of gas } (M) = \frac{6.00 \text{ g}}{0.0319 \text{ mol}} = 188 \text{ g mol}^{-1}$$

OR

2 points are earned for calculating the molar mass using $M = \frac{DRT}{P}$

$$M = \frac{DRT}{P} = \frac{6.00 \text{ g L}^{-1} \times 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1} \times 375 \text{ K}}{0.983 \text{ atm}}$$

$$= 188 \text{ g mol}^{-1}$$

- 1 point is earned for the molecular formula that is consistent with the molar mass calculated in part (b)(i).

Each CH₂Br unit has mass of 12.011 + 2(1.0079) + 79.90 = 93.9 g, and $\frac{188 \text{ g}}{93.9 \text{ g}} = 2.00$, so there must be two CH₂Br units per molecule. Therefore, the molecular formula of the compound is C₂H₄Br₂.



0	1	2	3
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The student response earns three of the following points:

- 1 point is earned for applying the gas law to calculate n .

$$PV = nRT \Rightarrow \frac{PV}{RT} = n$$

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$$\frac{(0.983 \text{ atm})(1.00 \text{ L})}{(0.0821 \text{ L atm mol}^{-1}\text{K}^{-1})(375 \text{ K})} = 0.0319$$

1 point is earned for calculating the molar mass.

$$\text{molar mass of gas } (M) = \frac{6.00 \text{ g}}{0.0319 \text{ mol}} = 188 \text{ g mol}^{-1}$$

OR

2 points are earned for calculating the molar mass using $M = \frac{DRT}{P}$

$$M = \frac{DRT}{P} = \frac{6.00 \text{ g L}^{-1} \times 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1} \times 375 \text{ K}}{0.983 \text{ atm}}$$

$$= 188 \text{ g mol}^{-1}$$

- 1 point is earned for the molecular formula that is consistent with the molar mass calculated in part (b)(i).

Each CH₂Br unit has mass of 12.011 + 2(1.0079) + 79.90 = 93.9 g, and $\frac{188 \text{ g}}{93.9 \text{ g}} = 2.00$, so there must be two CH₂Br units per molecule. Therefore, the molecular formula of the compound is C₂H₄Br₂.

Chapter 05 Part 3: Composition Analysis

41. Answer the following questions relating to gravimetric analysis.

In the first of two experiments, a student is assigned the task of determining the number of moles of water in one mole of $\text{MgCl}_2 \cdot n \text{H}_2\text{O}$. The student collects the data shown in the following table.

Mass of empty container	22.347 g
Initial mass of sample and container	25.825 g
Mass of sample and container after first heating	23.982 g
Mass of sample and container after second heating	23.976 g
Mass of sample and container after third heating	23.977 g

- (a) Explain why the student can correctly conclude that the hydrate was heated a sufficient number of times in the experiment.
- (b) Use the data above to
- calculate the total number of moles of water lost when the sample was heated, and
 - determine the formula of the hydrated compound.
- (c) A different student heats the hydrate in an uncovered crucible, and some of the solid spatters out of the crucible. This spattering will have what effect on the calculated mass of the water lost by the hydrate? Justify your answer.

In the second experiment, a student is given 2.94 g of a mixture containing anhydrous MgCl_2 and KNO_3 . To determine the percentage by mass of MgCl_2 in the mixture, the student uses excess $\text{AgNO}_3(aq)$ to precipitate the chloride ion as $\text{AgCl}(s)$.

- (d) Starting with the 2.94 g sample of the mixture dissolved in water, briefly describe the steps necessary to quantitatively determine the mass of the AgCl precipitate.
- (e) The student determines the mass of the AgCl precipitate to be 5.48 g. On the basis of this information, calculate each of the following.
- The number of moles of MgCl_2 in the original mixture
 - The percent by mass of MgCl_2 in the original mixture

Part A

1 point is earned for the correct explanation.

No additional mass was lost during the third heating, indicating that all the water of hydration had been driven off.

Chapter 05 Part 3: Composition Analysis



0	1
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The student response earns one of the following points:

1 point is earned for the correct explanation.

No additional mass was lost during the third heating, indicating that all the water of hydration had been driven off.

Part B (i)

1 point is earned for calculating the correct number of moles of water.

$$\text{mass of H}_2\text{O lost} = 25.825 - 23.977 = 1.848 \text{ g}$$

OR

$$25.825 - 23.976 = 1.849 \text{ g}$$

$$1.848 \text{ g H}_2\text{O} \times \frac{1 \text{ mol H}_2\text{O}}{18.02 \text{ g H}_2\text{O}} = 0.1026 \text{ mol H}_2\text{O}$$



0	1
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The student response earns one of the following points:

1 point is earned for calculating the correct number of moles of water.

$$\text{mass of H}_2\text{O lost} = 25.825 - 23.977 = 1.848 \text{ g}$$

OR

$$25.825 - 23.976 = 1.849 \text{ g}$$

$$1.848 \text{ g H}_2\text{O} \times \frac{1 \text{ mol H}_2\text{O}}{18.02 \text{ g H}_2\text{O}} = 0.1026 \text{ mol H}_2\text{O}$$

Part B (ii)

1 point is earned for calculating the correct number of moles of anhydrous MgCl_2 .

1 point is earned for writing the correct formula (with supporting calculations).

$$\text{mass of anhydrous MgCl}_2 = 23.977 - 22.347 = 1.630 \text{ g}$$

$$1.630 \text{ g MgCl}_2 \times \frac{1 \text{ mol MgCl}_2}{95.20 \text{ g MgCl}_2} = 0.01712 \text{ mol MgCl}_2$$

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$$\frac{0.1026 \text{ mol H}_2\text{O}}{0.01712 \text{ mol MgCl}_2} = 5.993 \approx 6 \text{ mol H}_2\text{O per mol MgCl}_2$$

⇒ formula is $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$



0	1	2
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The student response earns two of the following points:

1 point is earned for calculating the correct number of moles of anhydrous MgCl_2 .

1 point is earned for writing the correct formula (with supporting calculations).

mass of anhydrous $\text{MgCl}_2 = 23.977 - 22.347 = 1.630 \text{ g}$

$$1.630 \text{ g MgCl}_2 \times \frac{1 \text{ mol MgCl}_2}{95.20 \text{ g MgCl}_2} = 0.01712 \text{ mol MgCl}_2$$

$$\frac{0.1026 \text{ mol H}_2\text{O}}{0.01712 \text{ mol MgCl}_2} = 5.993 \approx 6 \text{ mol H}_2\text{O per mol MgCl}_2$$

⇒ formula is $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$

Part C

1 point point is earned for the correct answer with justification.

The calculated mass (or moles) of water lost by the hydrate will be too large because the mass of the solid that was lost will be assumed to be water when it actually included some MgCl_2 as well.



0	1
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The student response earns one of the following points:

1 point point is earned for the correct answer with justification.

The calculated mass (or moles) of water lost by the hydrate will be too large because the mass of the solid that was lost will be assumed to be water when it actually included some MgCl_2 as well.

Part D

2 points are point is earned for all three major steps: filtering the mixture, drying the precipitate, and determining the mass by difference.

1 point is earned for any two steps.

Add excess AgNO_3 .

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- Separate the AgCl precipitate (by filtration).
- Wash the precipitate and dry the precipitate completely.
- Determine the mass of AgCl by difference



0	1	2	3
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The student response earns three of the following points:

2 points are point is earned for all three major steps: filtering the mixture, drying the precipitate, and determining the mass by difference.

1 point is earned for any two steps.

Add excess AgNO₃.

- Separate the AgCl precipitate (by filtration).
- Wash the precipitate and dry the precipitate completely.
- Determine the mass of AgCl by difference

Part E (i)

1 point is earned for calculating the number of moles of AgCl. precipitate, and determining the mass by difference.

1 point is earned for conversion to moles of MgCl₂.

$$5.48 \text{ g AgCl} \times \frac{1 \text{ mol AgCl}}{143.32 \text{ g AgCl}} = 0.0382 \text{ mol AgCl}$$

$$0.0382 \text{ mol AgCl} \times \frac{1 \text{ mol Cl}}{1 \text{ mol AgCl}} \times \frac{1 \text{ mol MgCl}_2}{2 \text{ mol Cl}} = 0.0191 \text{ mol MgCl}_2$$



0	1	2
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The student response earns none of the following points:

1 point is earned for calculating the number of moles of AgCl. precipitate, and determining the mass by difference.

1 point is earned for conversion to moles of MgCl₂.

$$5.48 \text{ g AgCl} \times \frac{1 \text{ mol AgCl}}{143.32 \text{ g AgCl}} = 0.0382 \text{ mol AgCl}$$

$$0.0382 \text{ mol AgCl} \times \frac{1 \text{ mol Cl}}{1 \text{ mol AgCl}} \times \frac{1 \text{ mol MgCl}_2}{2 \text{ mol Cl}} = 0.0191 \text{ mol MgCl}_2$$

Chapter 05 Part 3: Composition Analysis**Part E (ii)**

1 point is earned for calculating the correct percentage.

$$0.0191 \text{ mol MgCl}_2 \times \frac{95.20 \text{ g MgCl}_2}{1 \text{ mol MgCl}_2} = 1.82 \text{ g MgCl}_2$$

$$\frac{1.82 \text{ g MgCl}_2}{2.94 \text{ g sample}} \times 100\% = 61.9\% \text{ MgCl}_2 \text{ by mass}$$



0

1

The student response earns one of the following points:

1 point is earned for calculating the correct percentage.

$$0.0191 \text{ mol MgCl}_2 \times \frac{95.20 \text{ g MgCl}_2}{1 \text{ mol MgCl}_2} = 1.82 \text{ g MgCl}_2$$

$$\frac{1.82 \text{ g MgCl}_2}{2.94 \text{ g sample}} \times 100\% = 61.9\% \text{ MgCl}_2 \text{ by mass}$$